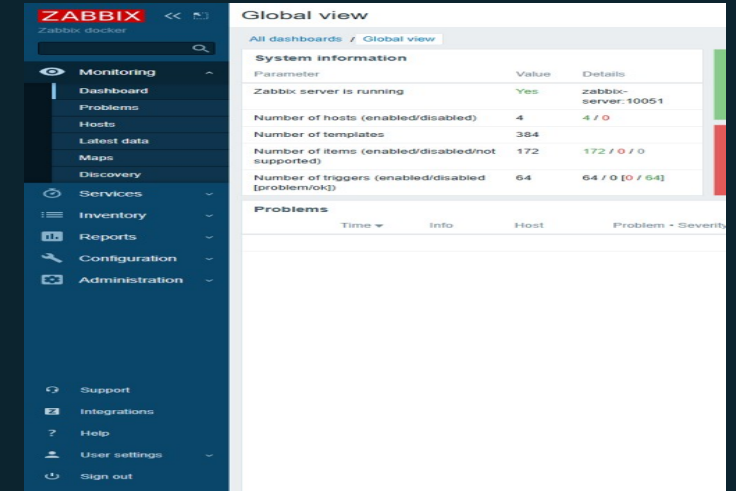


Proyecto 7

Monitoreo de infraestructura con Zabbix



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Problema

Servicios distribuidos pueden fallar sin aviso.

La revision manual tarda y no deja historial.

Se necesita visibilidad, alertas y evidencia para sustentacion.

Objetivo

Desplegar Zabbix 6.x en Docker Compose.

Monitorear web, base de datos, DNS y FTP.

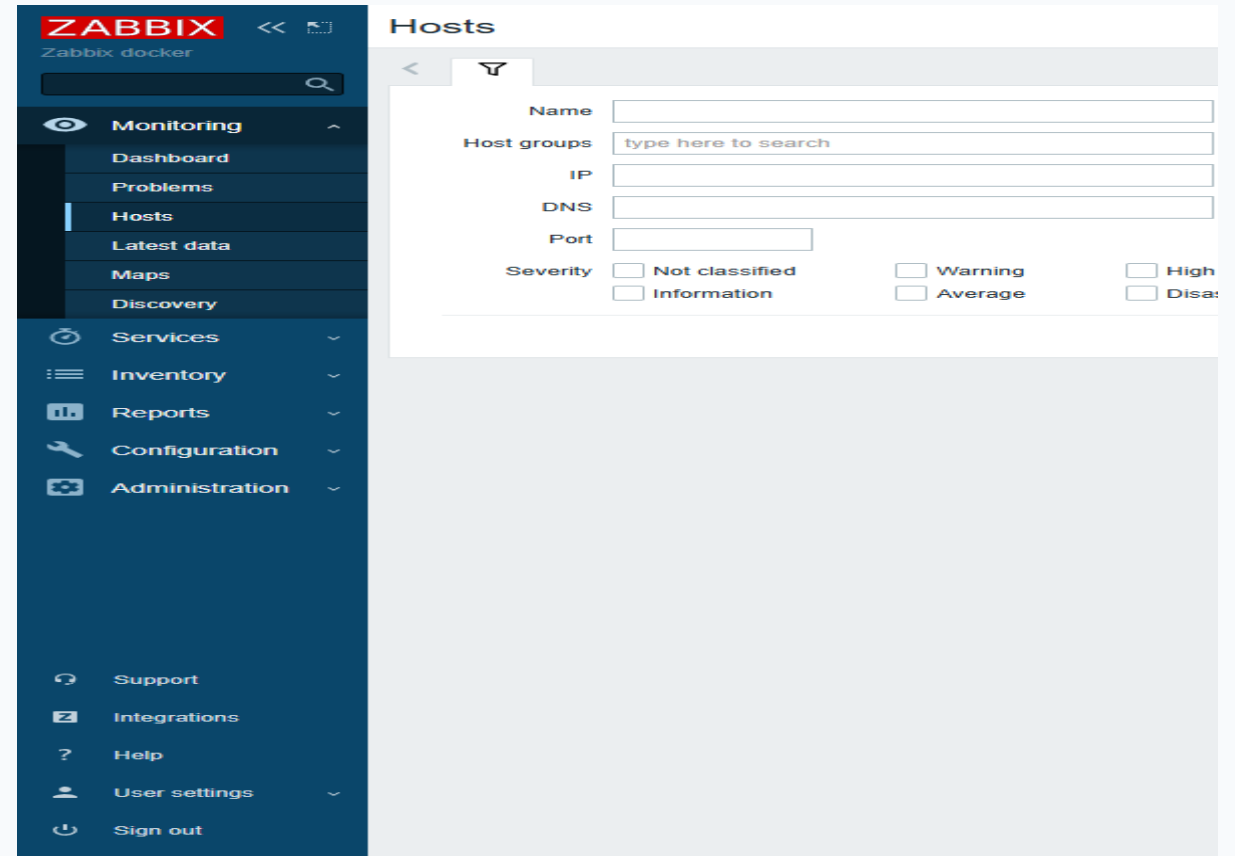
Validar triggers, dashboards, historicos y alertas por correo.

Arquitectura Docker

Zabbix Server + PostgreSQL + Zabbix Web.

MailHog simula SMTP para notificaciones.

Red interna proyecto7-monitoring para resolución por nombre.



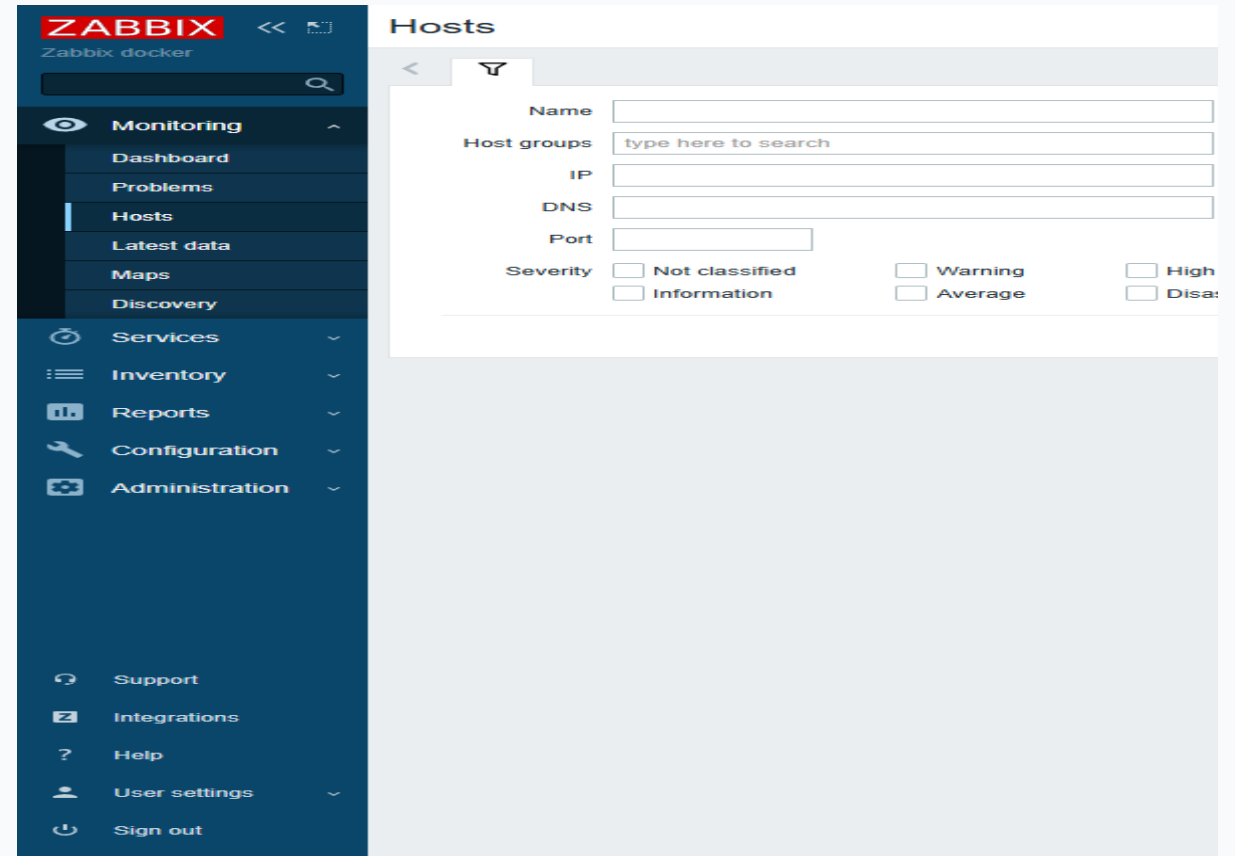
Inventario monitoreado

web-host: HTTP sobre Nginx.

db-host: MariaDB puerto 3306.

dns-host: CoreDNS puerto 53.

ftp-host: VSFTPD puerto 21.



The screenshot displays the Zabbix web interface. On the left is a dark blue sidebar with the Zabbix logo and a search bar. Below the search bar is a 'Monitoring' section with a sub-menu containing 'Dashboard', 'Problems', 'Hosts' (which is highlighted), 'Latest data', 'Maps', and 'Discovery'. Further down are sections for 'Services', 'Inventory', 'Reports', 'Configuration', and 'Administration'. At the bottom of the sidebar are links for 'Support', 'Integrations', 'Help', 'User settings', and 'Sign out'.

The main content area is titled 'Hosts' and contains a form for creating or editing a host. The form includes the following fields and options:

- Name:** A text input field.
- Host groups:** A text input field with the placeholder text 'type here to search'.
- IP:** A text input field.
- DNS:** A text input field.
- Port:** A text input field.
- Severity:** A set of radio buttons with the following options: 'Not classified', 'Warning', 'High', 'Information', 'Average', and 'Disa' (likely 'Disabled').

Implementacion

`docker-compose.yml` define todos los componentes.

`.env` centraliza credenciales y puertos.

`provision_zabbix.py` registra hosts, items y triggers por API.

Dashboard y datos

Latest data muestra disponibilidad y metricas.

Los checks de servicio entregan 1 cuando responden.

Los graficos permiten revisar comportamiento historico.

ZABBIX

Zabbix docker

Monitoring

Dashboard

Problems

Hosts

Latest data

Maps

Discovery

Services

Inventory

Reports

Configuration

Administration

Support

Integrations

Help

User settings

Sign out

Latest data

Host groups

type here to search

Hosts

type here to search

Name

Subfilter affects only filtered data

HOSTS

db-host 43

dns-host 43

ftp-host 43

web-host 43

TAGS

component 168

TAG VALUES

component: application 4

cpu 68

environment 4

memory 28

os 12

sec

DATA

With data

Without data

☐ Host	Name ▲
☐ dns-host	Disponibilidad DNS dns-service TCP
☐ ftp-host	Disponibilidad FTP ftp-service
☐ web-host	Disponibilidad HTTP web-service
☐ db-host	Disponibilidad MySQL db-service
☐ web-host	Linux: Available memory ?
☐ db-host	Linux: Available memory ?
☐ dns-host	Linux: Available memory ?
☐ ftp-host	Linux: Available memory ?
☐ web-host	Linux: Available memory in % ?

Prueba de caida

Se detiene web-service durante la demostracion.

Zabbix marca el trigger HTTP web-service no responde.

Al restaurar el contenedor, el evento queda resuelto.

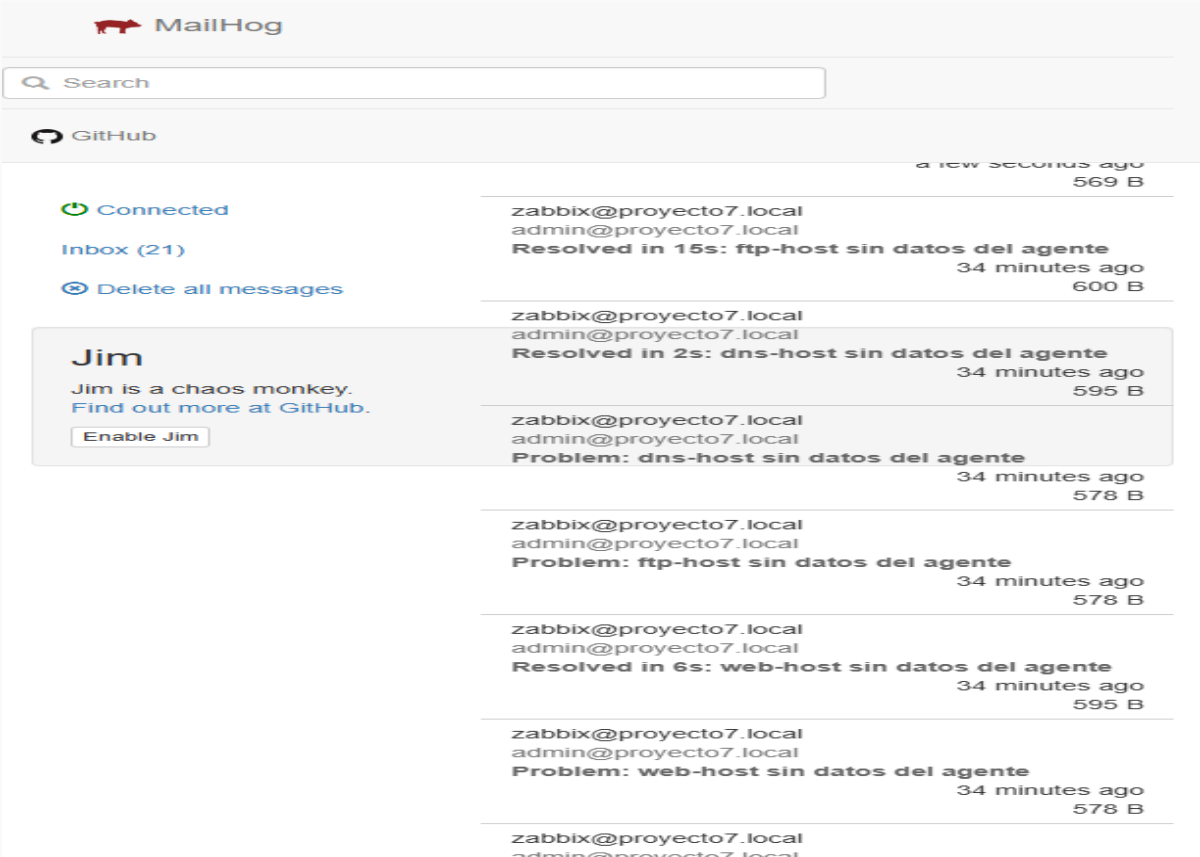
The screenshot shows the Zabbix web interface. The left sidebar has the Zabbix logo and a navigation menu with categories: Monitoring (Dashboard, Problems, Hosts, Latest data, Maps, Discovery), Services, Inventory, Reports, Configuration, and Administration. Below this are Support, Integrations, Help, User settings, and Sign out. The main content area is titled 'Problems' and includes tabs for 'Recent problems', 'Problems', and 'History'. There are search filters for Host groups, Hosts, and Triggers. A 'Problem' field is also present. Below these are checkboxes for Severity (Not classified, Warning, High, Information, Average, Disaster) and an 'Age less than' field set to 14 days. At the bottom, there is a table with columns: Time, Severity, Recovery time, Status, and Ir. The table contains one row with the value '00:06:16' in the Time column, 'High' in the Severity column, and 'PROBLEM' in the Status column. Below the table, it says '0 selected' and there is a 'Mass update' button.

Alertas con MailHog

MailHog recibe correos de problema y recuperacion.

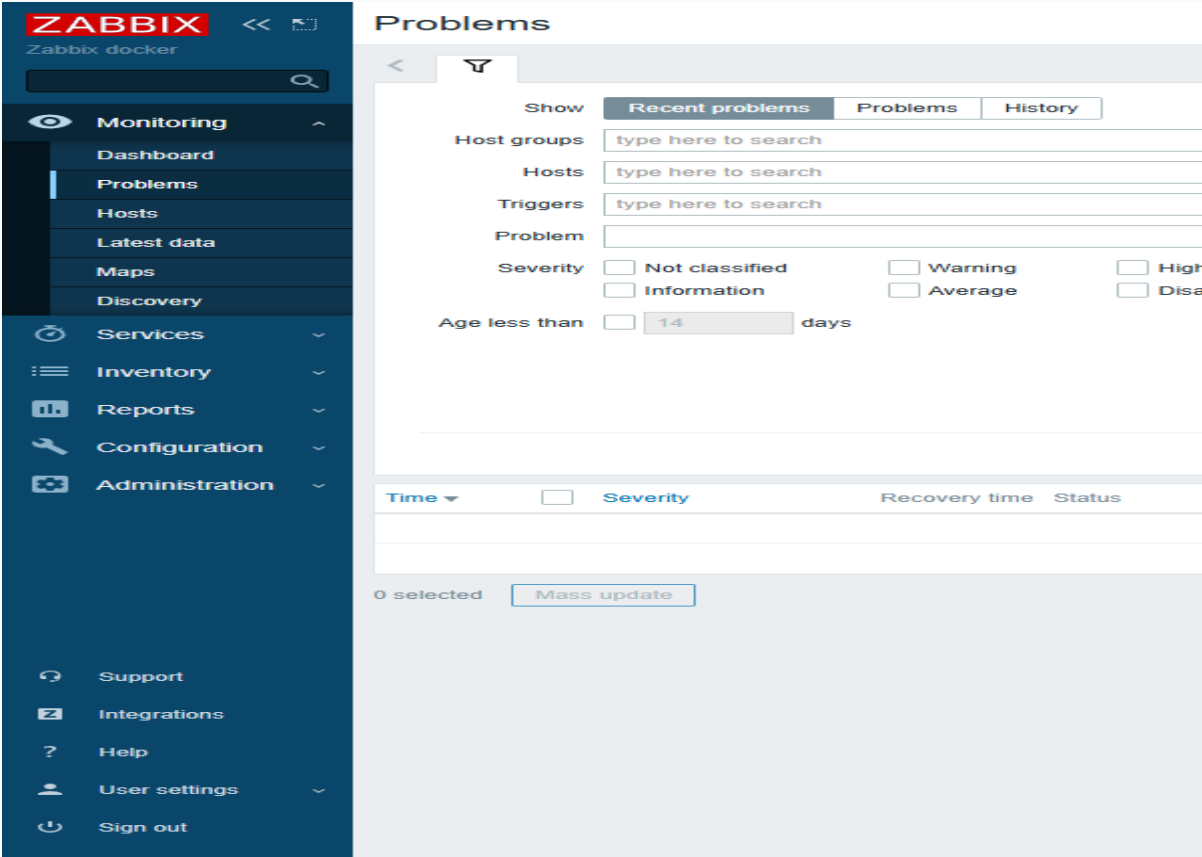
No se usan cuentas reales ni servidores externos.

La evidencia valida el flujo completo de notificacion.



Resultados

- Contenedores principales saludables.
- Cuatro servicios con check de disponibilidad activo.
- Alertas generadas y recuperadas durante la prueba.



Conclusiones

Zabbix centraliza observabilidad operativa.

Docker Compose hace el despliegue reproducible.

La solución cumple los requerimientos del Proyecto 7.

Demo en vivo

Abrir Zabbix: `http://localhost:8088`

Abrir MailHog: `http://localhost:8025`

Ejecutar: `.\scripts\test-failure.ps1 -Service web-service -Seconds 90`